Features

- Vin Range Up to 40V
- Output Range: 2.6V～5.0V
- Maximum Output Current: 250mA
- PSRR: 75dB@1KHz
- Dropout Voltage: 500mV@I_{OUT}=100mA
- Ultra Low Quiescent Current: 2μA Typ.
- Output Voltage Tolerances of ±2% Over the Temperature Range
- Internal Thermal Overload Protection
- Built-in Foldback Current Limit

Applications

- E-meters, Water Meters and Gas Meters
- Cellphones, Radiophone, Digital Cameras
- Fire Alarm, Smoke Detector
- Appliances and White Goods

General Description

The TX84251 is ultra-low quiescent current regulator features and low dropout voltage. With 2μA quiescent current at no load. The TX84251 retains all the features that are common to low dropout regulators including a low dropout PMOS pass device, foldback current limiter, and thermal shutdown.

The TX84251 has 40V maximum operating voltage, excellent load transient response, and -40°C to 125°C operating temperature range, and ±2% output voltage tolerance over the entire output current, input voltage, and temperature range.

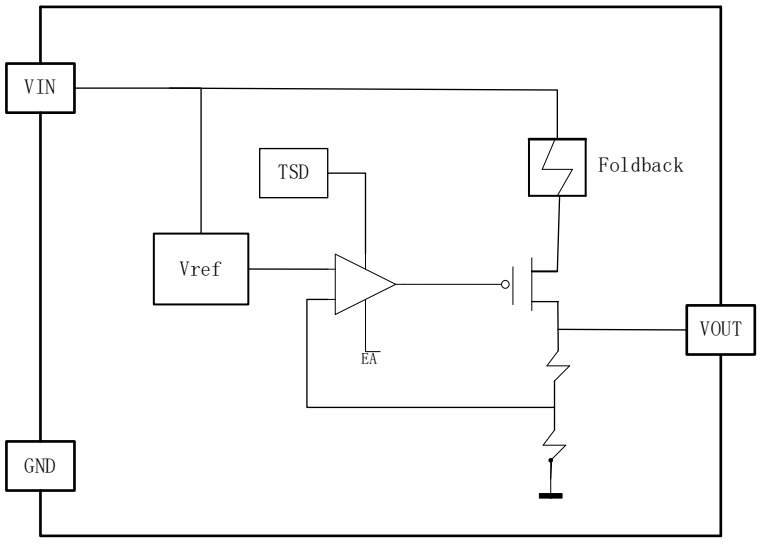
The TX84251 regulators are available in standard SOT23-3L, SOT89-3 and TO92 packages.

Order Information

TX84251-①②③④

Designator	Symbol	Description
①②	Integer	Output Voltage(2.6~5.0V)
③	M	Package: SOT23-3L
	P	Package: SOT89-3
	T	Package: TO92
④	R	Halogen Free

Block Diagram



Pin Assignment

SOT23-3 (Top View)

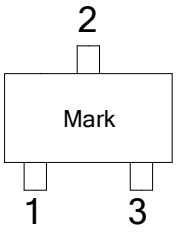


Table1：TX84251-XXMR series (SOT23-3L PKG)

PIN NO.	PIN NAME	FUNCTION
1	GND	GND pin
2	VIN	Input voltage pin
3	VOUT	Output voltage pin

SOT89 (Top View)

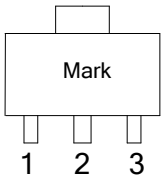


Table2：TX84251-XXPR series (SOT89-3 PKG)

PIN NO.	PIN NAME	FUNCTION
1	GND	GND pin
2	VIN	Input voltage pin
3	VOUT	Output voltage pin

TO92 (Top View)

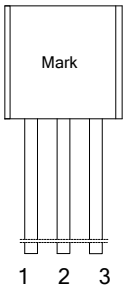


Table3：TX84251-XXTR series (TO92 PKG)

PIN NO.	PIN NAME	FUNCTION
1	GND	GND pin
2	VIN	Input voltage pin
3	VOUT	Output voltage pin

Absolute Maximum Ratings

Input Voltage……-0.3V to 45V
V_{EN}……-0.3V to VIN
V_{OUT}……-0.3V to 12V
ESD Rating……5.5kV
Storage Temperature ....-55℃ to 150℃
Lead Temperature(Soldering, 10 sec.). .....260℃
Operating Temperature…… -40℃ to 125℃

Note: These are stress ratings only. Stresses exceeding the range specified under “Absolute Maximum Ratings” may cause substantial damage to the device. Functional operation of this device at other conditions beyond those listed in the specification is not implied and prolonged exposure to extreme conditions may affect device reliability.

Electrical Characteristics

The following specifications apply for V_{IN}=12V, I_{OUT}=1mA, C_{IN}=C_{OUT}=10uF, T_J=25℃, unless specified otherwise.

SYMBOL	ITEMS	CONDITIONS	MIN	TYP	MAX	UNIT
V_{IN}	Input Voltage		2.7		40	V
V_{OUT}	Output Range		2.6		5.0	V
ΔV_{OUT}	Output Voltage Accuracy		-2	V_{OUT}	2	%
I_Q	Quiescent Current	V_{IN} = V_{OUT} + 1V	--	2	--	μA
V_{DROP}	Dropout Voltage	I_{OUT}=100mA	--	0.5	--	V
		I_{OUT}=250mA	--	1.3	--	
$\triangle V_{LINE}$	Line Regulation	V_{IN} = V_{OUT} + 1V to 40V, or V_{IN} = 5V to 40V, if V_{OUT} < 4V	--	30	--	mV
$\triangle V_{LOAD}$	Load Regulation	I_{OUT} = 1mA to 100mA	--	50	--	mV
PSRR	Power Supply Rejection Rate VIN=VOUT+1V Ripple=0.2Vp-p IOUT=10mA	Freq.=1kHz	--	75	--	dB
I_{SHORT}	I_{OUT} @ V_{OUT} Short to Ground	VIN=6V, VOUT=0V	--	50	--	mA

Note1: The parameters of Cin and Cout must be greater than 1uF.

Application Circuits

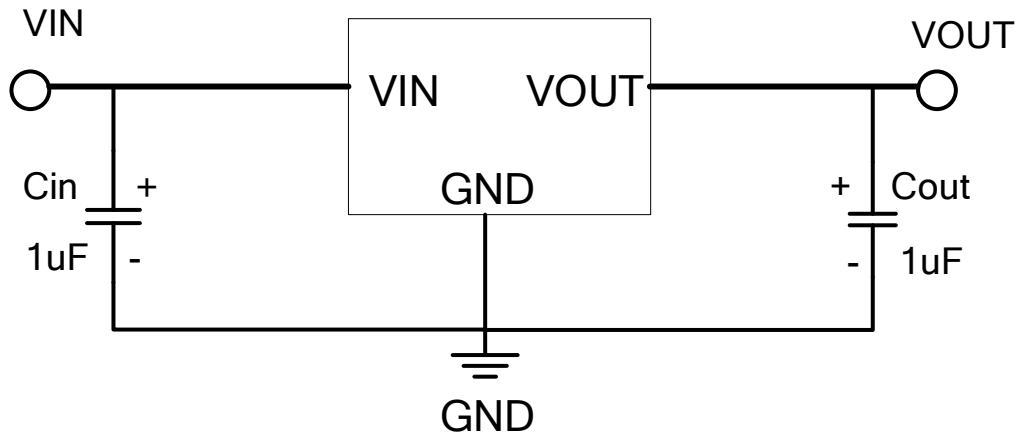


Figure1

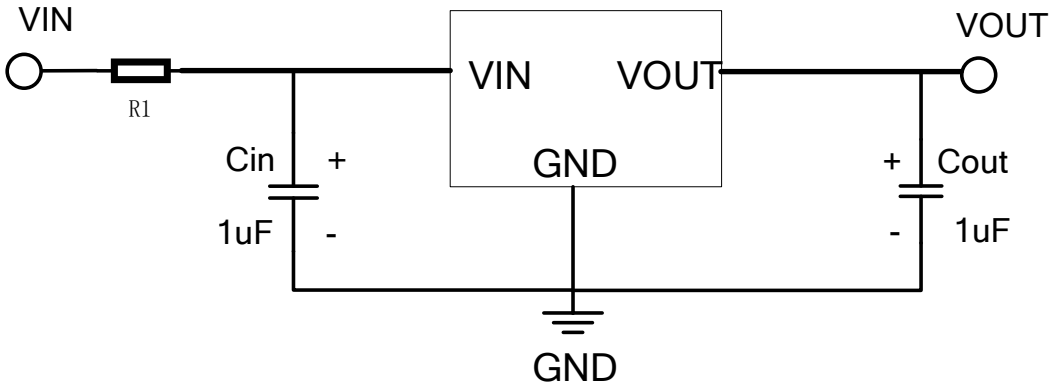


Figure2

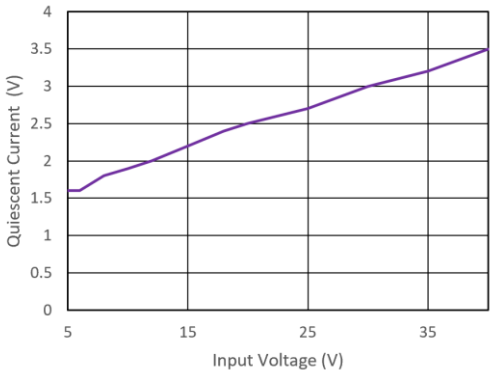
Note1: The parameters of Cin and Cout must be greater than 1uF.

Note2: If VIN＞18V，you'd better add a resistor to the input. The resistance value of R1 should be set to 0.25Ω.

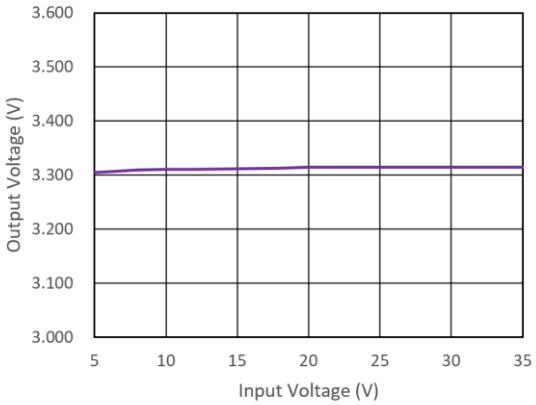
Typical Performance Characteristics

C_{IN}=10uF, C_{OUT}=10uF, T_J=25°C, unless specified otherwise

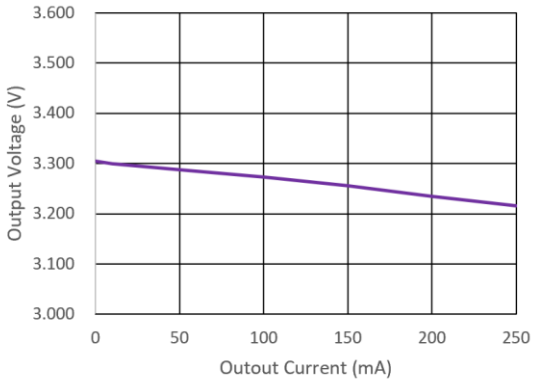
(1) Quiescent Current vs Input Voltage



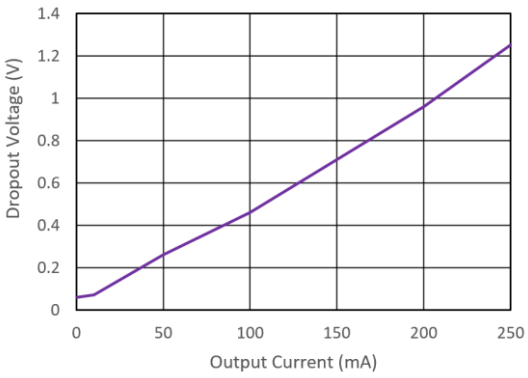
(2) Output Voltage vs Input Voltage



(3) Output Voltage vs Output Current

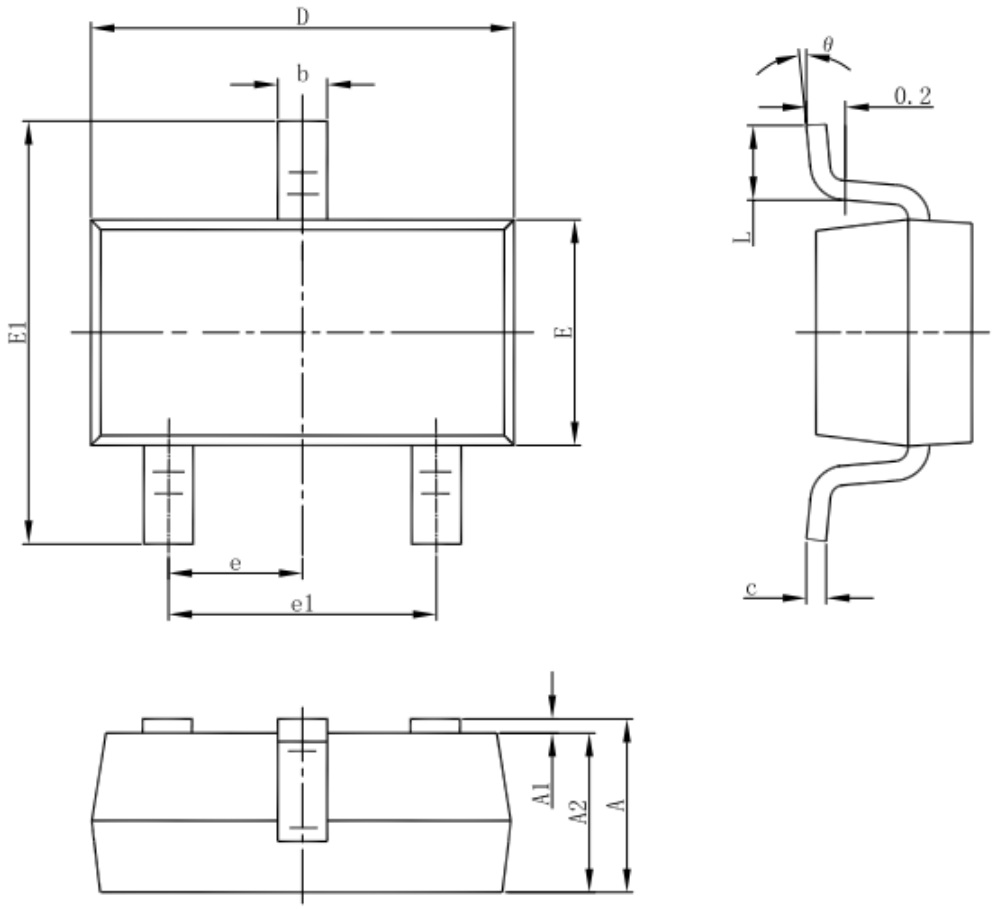


(4) Dropout Voltage vs Output Current



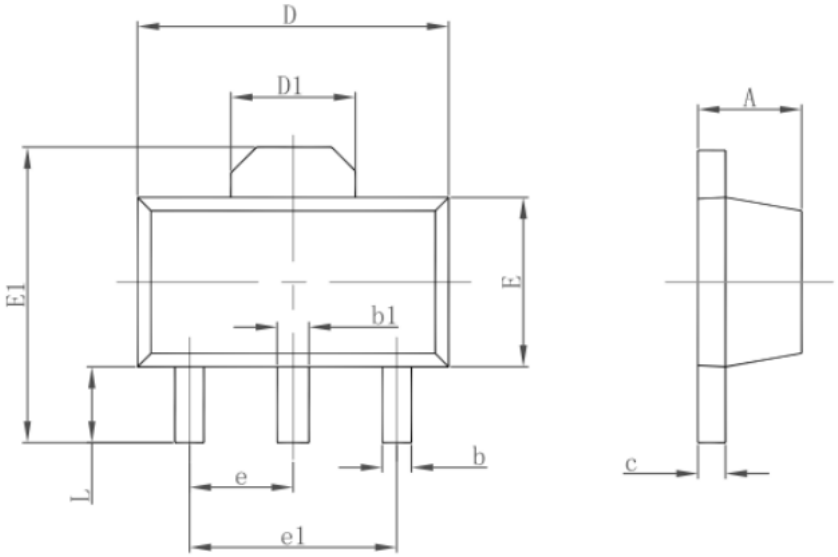
Package Information

3-pin SOT23-3 Outline Dimensions



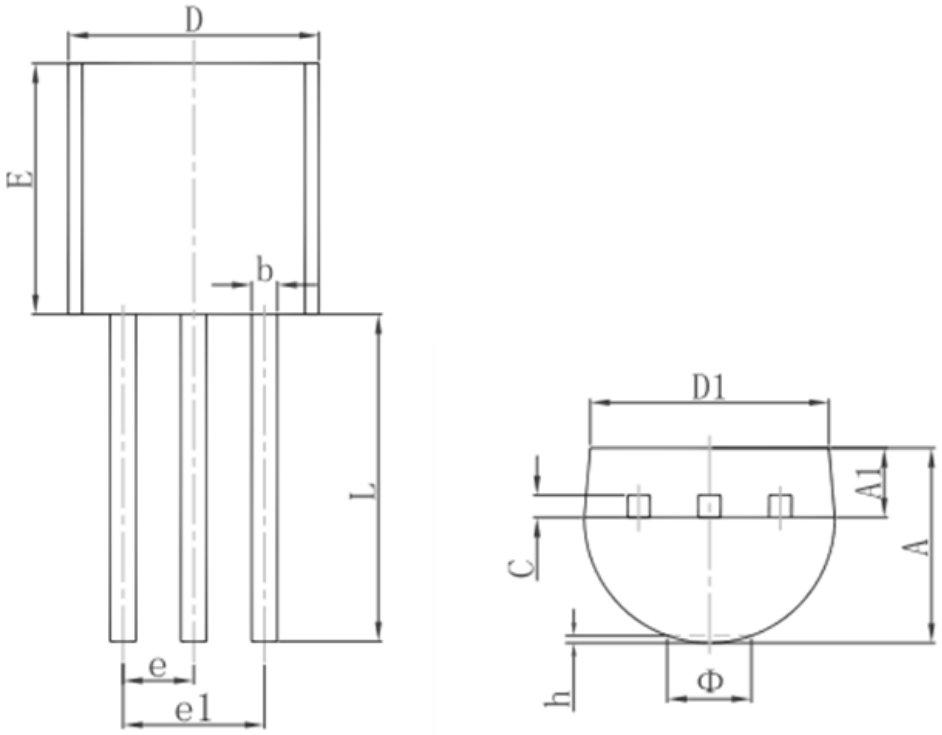
Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min	Max	Min	Max
A	1.050	1.250	0.041	0.049
A1	0.000	0.100	0.000	0.004
A2	1.050	1.150	0.041	0.045
b	0.300	0.500	0.012	0.020
c	0.100	0.200	0.004	0.008
D	2.820	3.020	0.111	0.119
E	1.500	1.700	0.059	0.067
E1	2.650	2.950	0.104	0.116
e	0.950(BSC)		0.037(BSC)	
e1	1.800	2.000	0.071	0.079
L	0.300	0.600	0.012	0.024
θ	0°	8°	0°	8°

SOT89-3L Outline Dimensions



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	1.400	1.600	0.055	0.063
b	0.320	0.520	0.013	0.020
b1	0.400	0.580	0.016	0.023
c	0.350	0.440	0.014	0.017
D	4.400	4.600	0.173	0.181
D1	1.550 REF.		0.061 REF.	
E	2.300	2.600	0.091	0.102
E1	3.940	4.250	0.155	0.167
e	1.500 TYP.		0.060 TYP.	
e1	3.000 TYP.		0.118 TYP.	
L	0.900	1.200	0.035	0.047

3-pin TO92 Outline Dimensions



Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	3.300	3.700	0.130	0.146
A1	1.100	1.400	0.043	0.055
b	0.380	0.550	0.015	0.022
c	0.360	0.510	0.014	0.020
D	4.300	4.700	0.169	0.185
D1	3.430		0.135	
E	4.300	4.700	0.169	0.185
e	1.270 TYP.		0.050 TYP.	
e1	2.440	2.640	0.096	0.104
L	14.100	14.500	0.555	0.571
Φ		1.600		0.063
h	0.000	0.380	0.000	0.015

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